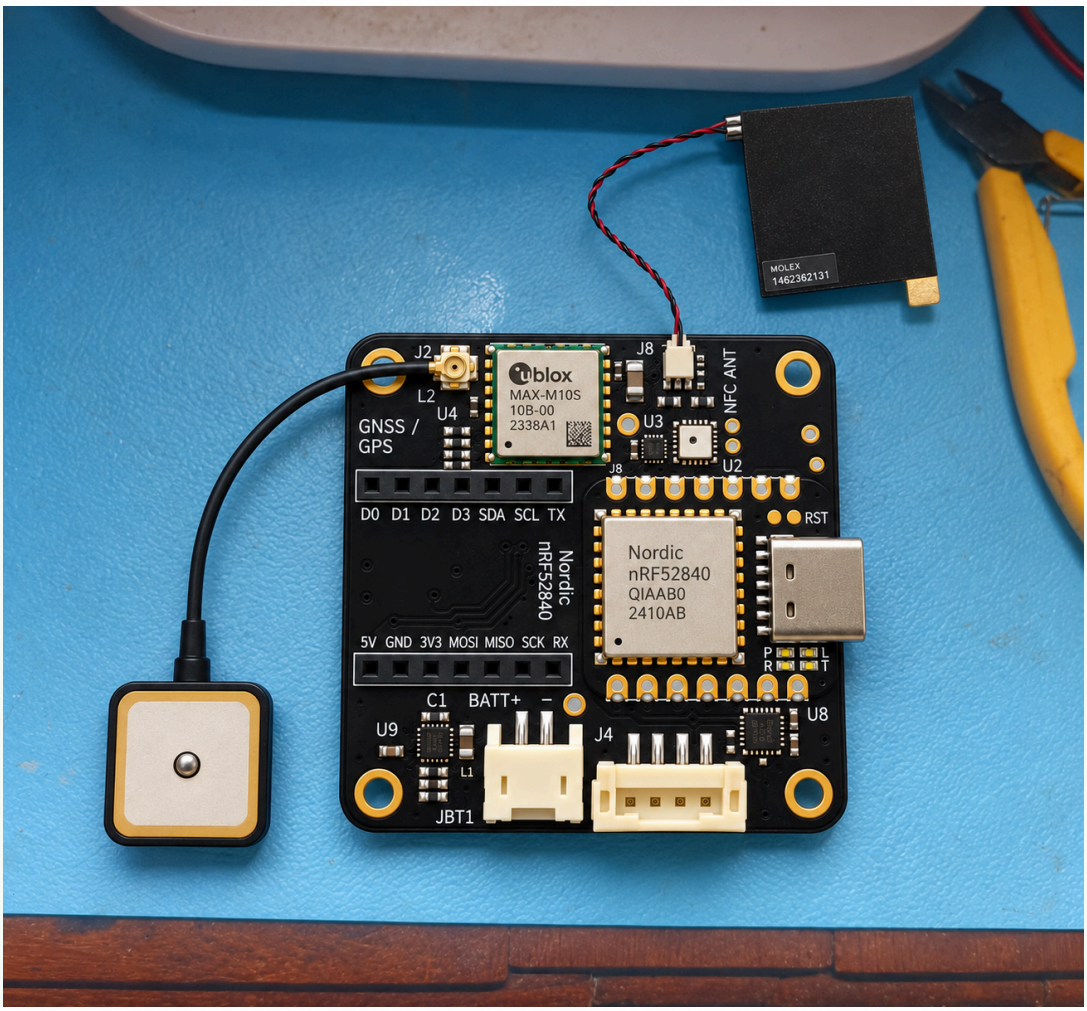
EMBEDDED SYSTEMS PORTFOLIO

Nordic BLE Hardware & Firmware Engineering

Hardware Design • Embedded Firmware • Wireless • Manufacturing • Validation

End-to-end development of connected embedded systems - from circuit architecture and PCB design through Nordic firmware, wireless connectivity, production handoff and physical validation.



GNSS
External active antenna
Positioning / RF

NFC
External loop antenna
Tap / proximity interface

LiPo POWER
1-cell battery input
Portable low-power operation

COMPLETE EMBEDDED ENGINEERING CAPABILITY

SCHEMATIC + PCB	ZEPHYR / NCS	BLE + RF	BOM / CPL / GERBER	PHYSICAL VALIDATION
Design & implementation	Embedded firmware	Custom GATT & wireless	Manufacturing handoff	Real hardware testing

nRF52840 | KiCad | Embedded C | Zephyr RTOS | Nordic nRF Connect SDK | BLE GATT | Sensors | GNSS / RF

Engineering evidence and detailed project work follow.

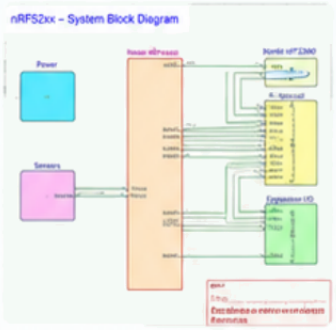
NORDIC BLE HARDWARE & FIRMWARE ENGINEERING

From System Concept to Working Connected Hardware

Hardware architecture, schematic design, PCB implementation, manufacturing handoff, embedded firmware and physical system validation.

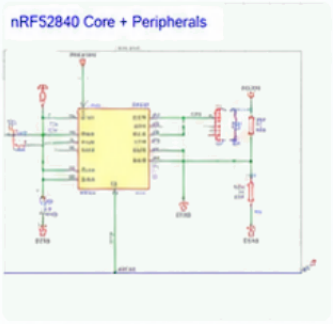
IDEA → ARCHITECTURE → SCHEMATIC → PCB → MANUFACTURING → FIRMWARE → VALIDATION → HARDWARE

1 SYSTEM ARCHITECTURE



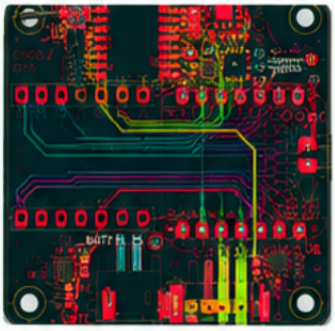
Functional subsystems and interface definitions.

2 SCHEMATIC DESIGN



Power, sensing, GNSS and sensor circuits.

3 PCB IMPLEMENTATION



Four-layer placement, routing and access.

4 MANUFACTURING



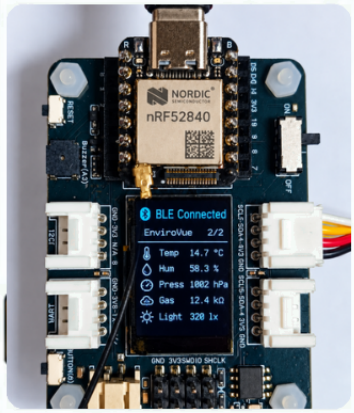
Gerber, BOM and CPL assembly workflow.

5 EMBEDDED FIRMWARE



Zephyr / NCS firmware development evidence.

6 PHYSICAL VALIDATION



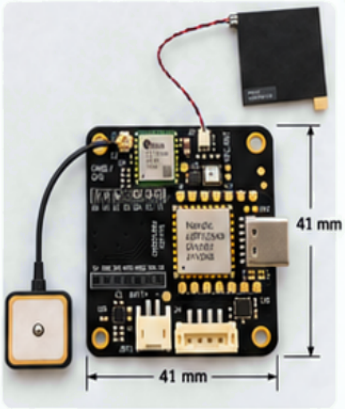
Hardware build and live validation on Nordic platform.

7 APPLICATION INTEGRATION



EnviroVue application interface evidence.

8 COMPACT HARDWARE



41 × 41 mm PCB with GNSS and expansion.

END-TO-END ENGINEERING

CONCEPT > SCHEMATIC > PCB > MANUFACTURING > FIRMWARE > VALIDATION > HARDWARE

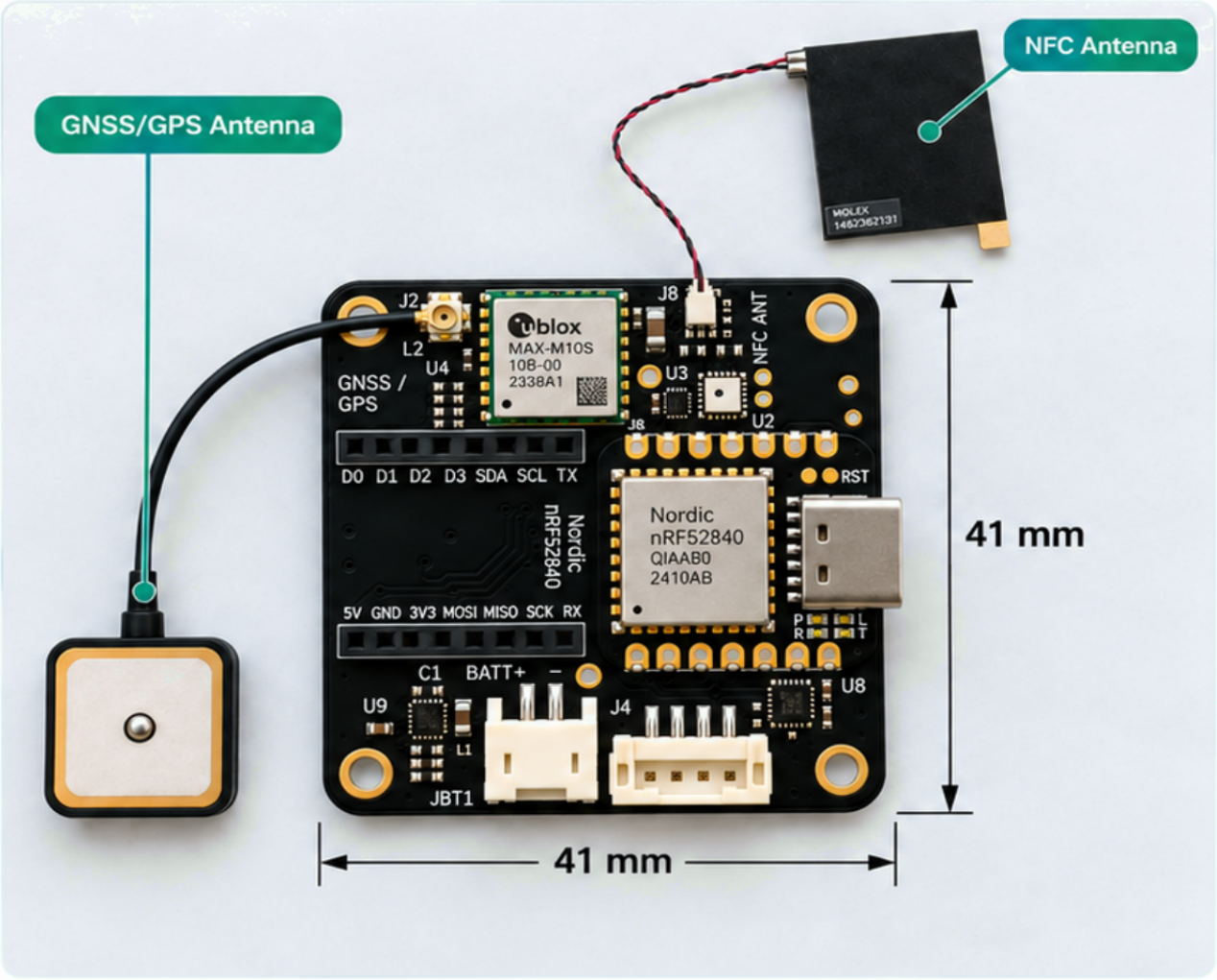
nRF52840 | KiCad | Embedded C | Zephyr RTOS | Nordic nRF Connect SDK | BLE GATT | Sensors | GNSS / RF

Engineering process overview - detailed evidence is provided throughout the portfolio.

CUSTOM nRF52840 HARDWARE DESIGN

Compact Nordic BLE with GNSS/GPS and NFC

Custom KiCad design combining wireless, sensing, positioning, modular expansion and low-power system architecture.



ENGINEERING FEATURE SET

BLE + NFC + DEBUG
nRF52840 BLE core with dedicated NFC1/NFC2 connections and SWD programming/debug.

POWER ARCHITECTURE
1-cell LiPo system with TPS63802 buck-boost generating a +3V3_SYS rail, plus power test access.

MOTION / CONDITION
ADXL372 high g motion and condition sensing over SPI with interrupt signalling.

ENVIRONMENTAL + LIGHT
BME688 environmental sensing plus VEML6035 ambient-light sensing on I2C.

GNSS + ACTIVE ANTENNA
MAX-M10S GNSS with UART/PPS, external RF connection and VCC_RF active-antenna bias.

MODULAR EXPANSION
I2C, SPI, UART and GPIO support external sensors, displays, communications and additional wireless interfaces, including LoRa-class radios. Dedicated +3V3_SYS expansion supply.

KICAD / DFT
Hierarchical design: Power/USB -> Core -> Motion -> Environmental -> Expansion -> GNSS. SWD, test access and manufacturing oriented design checks.

RF / COMPLIANCE: Pre-certified nRF52840 radio module (FCC / CE) provides a strong compliance foundation; final product compliance remains application dependent.

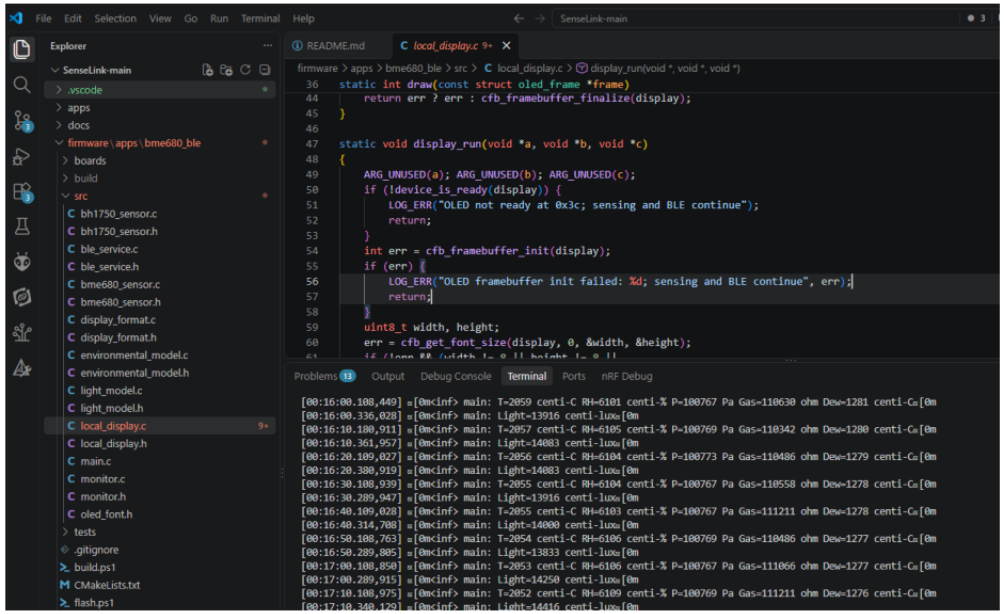
nRF52840 BLE + NFC | LiPo + Buck-Boost | ADXL372 | BME688 + VEML6035 | MAX-M10S + Active Antenna | Modular I2C / SPI / UART / GPIO | SWD / DFT

Custom hardware-design evidence: architecture, schematics capture, PCB implementation, test access and manufacturing preparation.

NORDIC FIRMWARE ENGINEERING

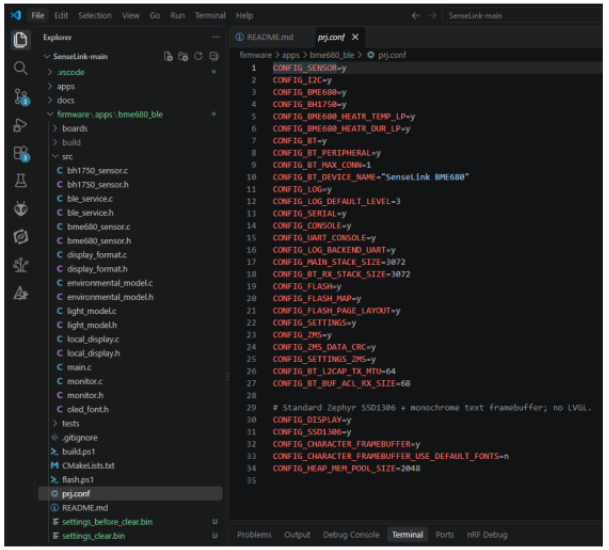
Zephyr RTOS • nRF Connect SDK • Real-Hardware Validation

Modular embedded C firmware developed and exercised on physical Nordic hardware with live sensor telemetry and Bluetooth LE.



MODULAR SOURCE + LIVE TELEMETRY

Separate sensor, BLE, display and application modules with live environmental readings from physical hardware.



ZEPHYR CONFIGURATION

Sensors • I²C • BLE peripheral • logging • flash/settings • SSD1306

MODULAR EMBEDDED C

Dedicated modules for BME680, BH1750, BLE services, environmental/light models, display formatting, monitoring and application control.

ZEPHYR / NCS

Zephyr sensor and I²C frameworks, Bluetooth LE peripheral configuration, UART logging, display support and explicit memory/transport configuration.

PERSISTENT SETTINGS

Flash map/page layout with Zephyr Settings and ZMS storage, including data CRC support for non-volatile application configuration.

PHYSICAL VALIDATION

Firmware built, flashed and exercised on real Nordic hardware; runtime logs show live temperature, humidity, pressure, gas and light telemetry.

SENSORS → MODULAR DRIVERS → APPLICATION LOGIC → BLE → LIVE TELEMETRY

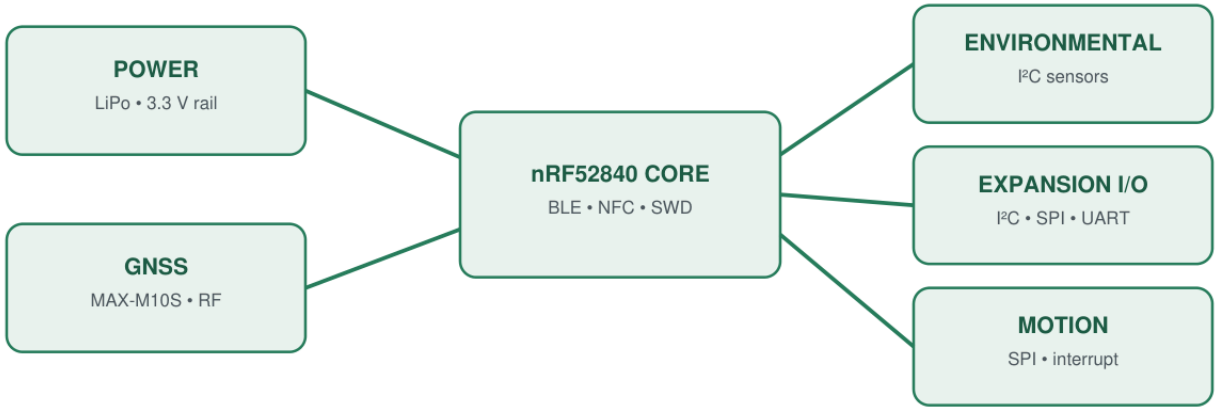
Embedded C | Zephyr RTOS | Nordic nRF Connect SDK | I²C | BLE | Logging | Flash / Settings | Hardware Debug

SCHEMATIC & ELECTRONIC DESIGN

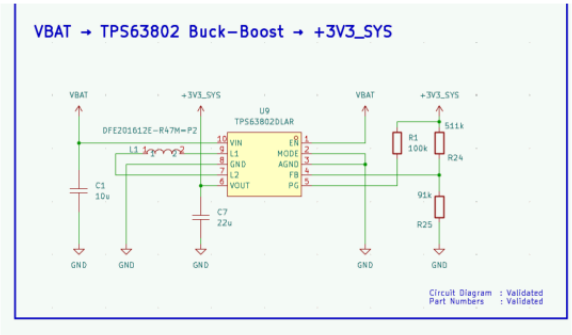
System Architecture • Power • GNSS / RF

Hierarchical circuit design translating system requirements into defined power, sensing, positioning and interface subsystems.

SYSTEM-LEVEL SCHEMATIC ARCHITECTURE



Functional partitioning keeps power, sensors, positioning and external interfaces explicit and reviewable.



POWER ARCHITECTURE



GNSS / ACTIVE-ANTENNA RF INTERFACE

SYSTEM DECOMPOSITION

Hierarchical design separates power, Nordic core, motion, environmental sensing, expansion I/O and GNSS into functional blocks with defined interfaces.

POWER DESIGN

Single-cell battery input is converted to the 3.3 V system rail using a TPS63802 buck-boost stage with external inductor, input/output capacitance and feedback network.

GNSS / RF

MAX-M10S positioning interface includes UART and time-pulse signals plus an external coaxial RF connection. VCC_RF feeds the antenna line through a 27 nH RF choke for active-antenna bias.

DESIGN FOR DEBUG

Dedicated programming/debug access and exposed subsystem interfaces support board bring-up, firmware development and engineering verification.

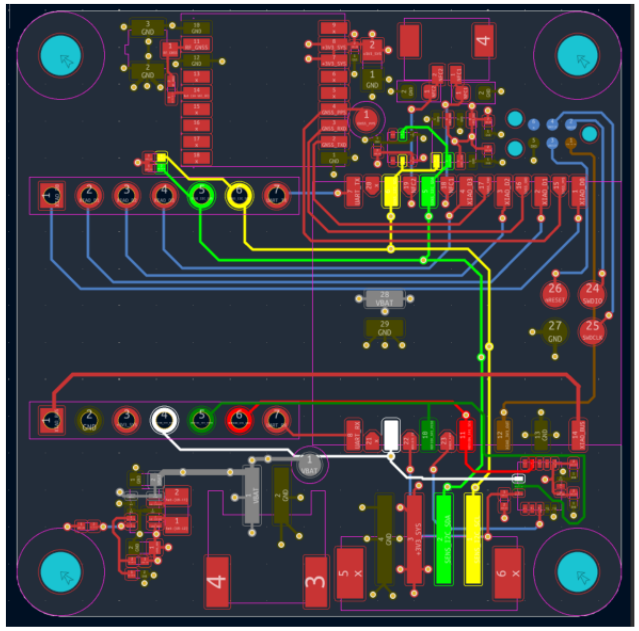
REQUIREMENTS -> SYSTEM ARCHITECTURE -> CIRCUIT DESIGN -> INTERFACE DEFINITION -> PCB IMPLEMENTATION

KiCad | nRF52840 | Power Architecture | I2C / SPI / UART | GNSS / RF | SWD / Debug | Modular Expansion

PCB LAYOUT & MANUFACTURING ENGINEERING

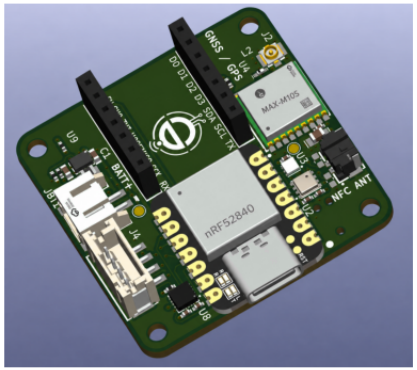
Routed PCB • Placement • DFT • Manufacturing Handoff

Custom Nordic PCB implementation in KiCad, carrying the schematic architecture through placement, routing and fabrication-ready outputs.



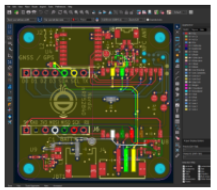
ROUTED PCB - COPPER, VIAS & INTERCONNECT

Clean routing view exposes the implemented signal paths, connector fan-out, debug access and subsystem placement.



3D IMPLEMENTATION

Populated-board integration with nRF52840 core and MAX-M10S GNSS module.



KICAD PCB EDITOR WORKFLOW

PLACEMENT & PARTITIONING

Functional areas are physically organized around the Nordic core, GNSS/RF section, power conversion, sensor interfaces, connectors and expansion access.

ROUTING & FAN-OUT

Implemented copper shows signal breakout from the core to external I/O, GNSS, debug and peripheral interfaces, with vias and layer transitions visible in the routed design.

BRING-UP / DFT

Dedicated SWD/reset access, exposed rails and accessible connectors support programming, board bring-up, firmware development and bench verification.

FABRICATION HANDOFF

The KiCad design is carried through to manufacturing-oriented outputs including BOM, component-position data and Gerber/drill fabrication files.

SCHEMATIC -> PLACEMENT -> ROUTING -> DFT / BRING-UP ACCESS -> BOM / CPL / GERBERS

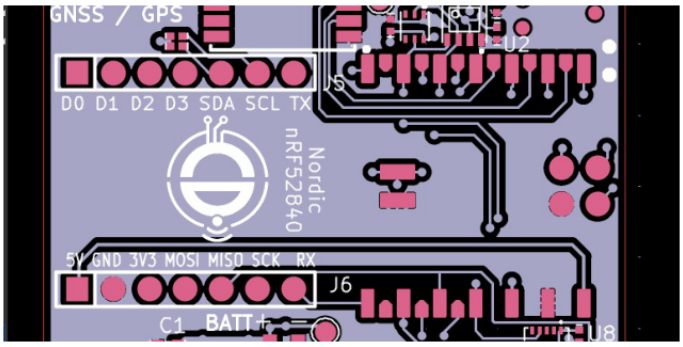
KiCad PCB Editor | Mixed-Signal Layout | GNSS / RF Integration | SWD / Debug | Manufacturing Outputs

MANUFACTURING & PRODUCTION HANDOFF

Assembly-Ready Manufacturing Data

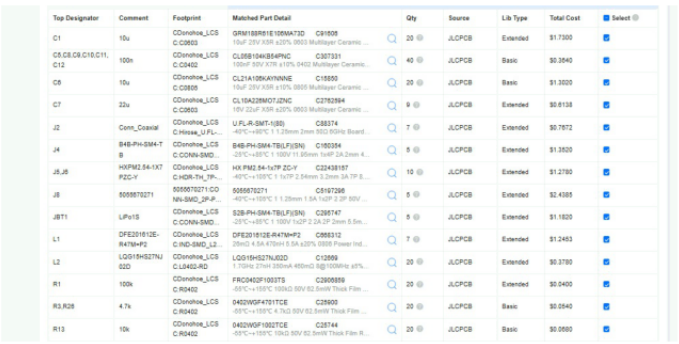
Gerbers • BOM • CPL • Assembly Review • Production Handoff

PCB DESIGN -> GERBER / DRILL -> BOM -> CPL -> ASSEMBLY REVIEW -> MANUFACTURING PACK



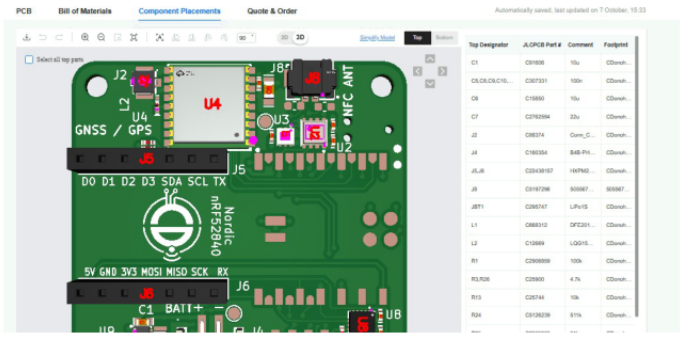
FABRICATION DATA

Gerber Viewer confirms fabrication layers and drill/pad data.



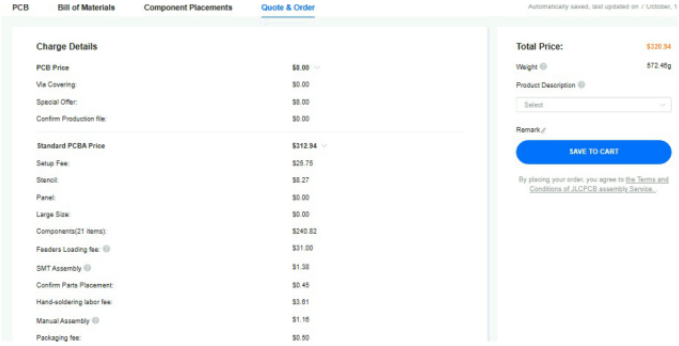
BOM & SOURCING VALIDATION

JLCPCB workflow maps designators, footprints and sourced parts.



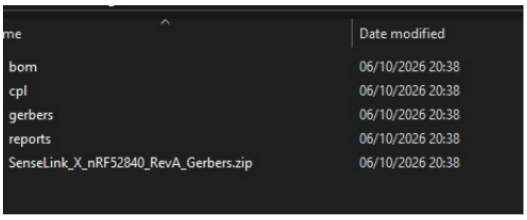
PLACEMENT / CPL REVIEW

JLCPCB interprets pick-and-place data in the assembly preview.



ASSEMBLY WORKFLOW

JLCPCB PCB + SMT data accepted through quotation stage.



ORGANIZED MANUFACTURING DELIVERABLE

Structured release pack separates BOM, CPL, Gerbers and reports, with a packaged Gerber archive ready for fabrication and assembly handoff.

BOM | CPL | GERBERS | REPORTS | FABRICATION ZIP

DESIGN -> FABRICATION DATA -> COMPONENT SOURCING -> PLACEMENT -> ASSEMBLY HANDOFF

KiCad | Gerber / Drill | BOM | CPL / Pick & Place | Component Sourcing | SMT Assembly

BLUETOOTH LE ENGINEERING

Custom GATT • Notifications • Device Configuration

Custom Zephyr/NCS Bluetooth LE interface carrying live environmental data and writable device configuration.

```c
CCC_CALLBACK(humidity)
CCC_CALLBACK(pressure)
CCC_CALLBACK(gas)
CCC_CALLBACK(dew)
CCC_CALLBACK(record)
CCC_CALLBACK(light)

#define MEASUREMENT(uuid, index, callback) \
    BT_GATT_CHARACTERISTIC(uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY, \
        BT_GATT_PERM_READ, read_measurement, NULL, (void *)&measurement_indices[index]), \
    BT_GATT_CCC(callback, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE)

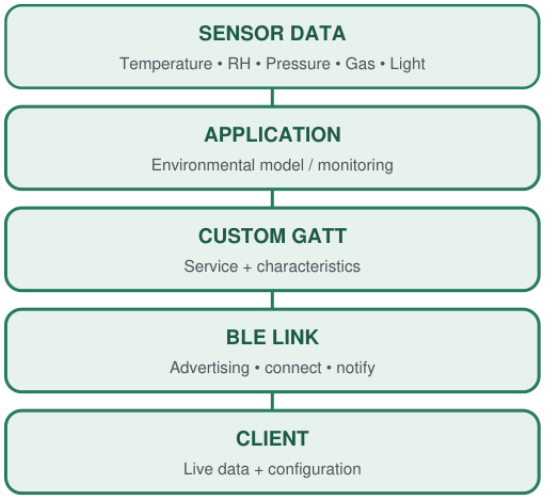
BT_GATT_SERVICE_DEFINE(environmental_service,
    BT_GATT_PRIMARY_SERVICE(&service_uuid),
    MEASUREMENT(&temperature_uuid.uuid, 0, temperature_ccc),
    MEASUREMENT(&humidity_uuid.uuid, 1, humidity_ccc),
    MEASUREMENT(&pressure_uuid.uuid, 2, pressure_ccc),
    MEASUREMENT(&gas_uuid.uuid, 3, gas_ccc),
    MEASUREMENT(&dew_uuid.uuid, 4, dew_ccc),
    BT_GATT_CHARACTERISTIC(&record_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY,
        BT_GATT_PERM_READ, read_record, NULL, NULL),
    BT_GATT_CCC(record_ccc, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE),
    BT_GATT_CHARACTERISTIC(&interval_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_WRITE,
        BT_GATT_PERM_READ | BT_GATT_PERM_WRITE, read_interval, write_interval, NULL),
    BT_GATT_CHARACTERISTIC(&light_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY,
        BT_GATT_PERM_READ, read_light, NULL, NULL),
    BT_GATT_CCC(light_ccc, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE)
);

static const struct bt_data advertising[] = {
    BT_DATA_BYTES(BT_DATA_FLAGS, BT_LE_AD_GENERAL | BT_LE_AD_NO_BREDR),
    BT_DATA_BYTES(BT_DATA_UUID128_ALL, ENV_UUID(0x8f3a0000)),
};
```

CUSTOM GATT SERVICE - ZEPHYR / NCS

Actual implementation: environmental characteristics, READ/NOTIFY, CCC subscriptions, writable interval and advertising.

BLE DATA PATH



CUSTOM GATT

Custom 128-bit service with dedicated environmental characteristics rather than relying only on a standard BLE profile.

READ + NOTIFY

Temperature, humidity, pressure, gas resistance, dew point and light are exposed for client reads and live notification updates.

TWO-WAY CONTROL

A writable interval characteristic lets the BLE client change the device sampling/update interval, with validation handled in firmware.

ROBUST BLE FLOW

Advertising, connection state, CCC subscriptions and notification delivery are integrated into the application architecture.

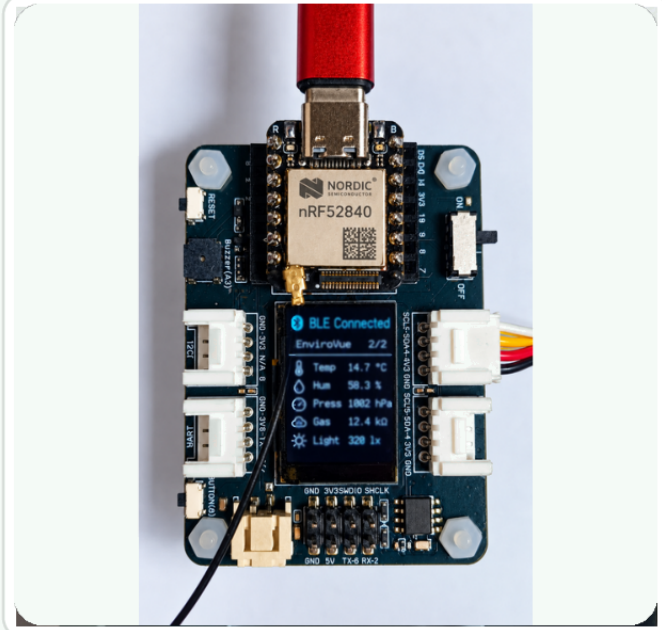
PHYSICAL VALIDATION | Nordic hardware -> Zephyr/NCS -> custom GATT -> BLE client -> live sensor data + device configuration

Embedded C | Zephyr Bluetooth Stack | Custom 128-bit UUIDs | GATT | CCC / Notifications | Advertising

END-TO-END SYSTEM VALIDATION

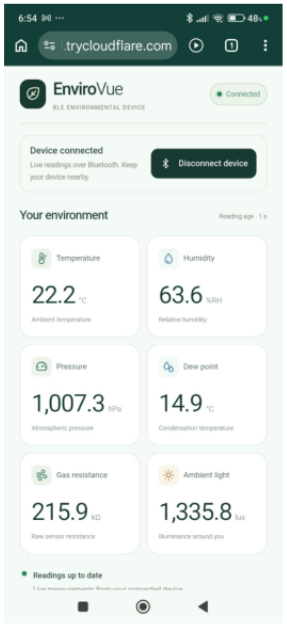
Physical Sensors • Nordic Firmware • BLE • Live Application

Working sensor-to-application chain validated on physical Nordic hardware with live telemetry and bidirectional BLE configuration.



NORDIC DEVELOPMENT BOARD

OLED shows BLE connection and live sensor values.



ENVIROVUE - LIVE BLE DASHBOARD

Connected client displaying six live environmental measurements.



LIVE TREND + DEVICE CONFIGURATION

Client changes update interval over BLE; device retains the setting.

PHYSICAL VALIDATION

Firmware was built, flashed and exercised on real Nordic hardware rather than simulator-only development.

LIVE SENSOR CHAIN

Environmental and ambient-light measurements are acquired on-device and presented locally before transport over BLE.

BIDIRECTIONAL BLE

Live measurements flow to the client while a writable GATT characteristic allows the application to change the device update interval.

APPLICATION INTEGRATION

EnviroVue demonstrates the complete interface: connection state, live values, recent trends and persistent device configuration.

BME680 + BH1750 -> PHYSICAL NORDIC HARDWARE -> ZEPHYR/NCS -> CUSTOM BLE GATT -> ENVIROVUE

Measured on hardware | Live BLE telemetry | Browser client | Recent-value trends | Writable persistent configuration